



THE UNIVERSITY OF
MELBOURNE

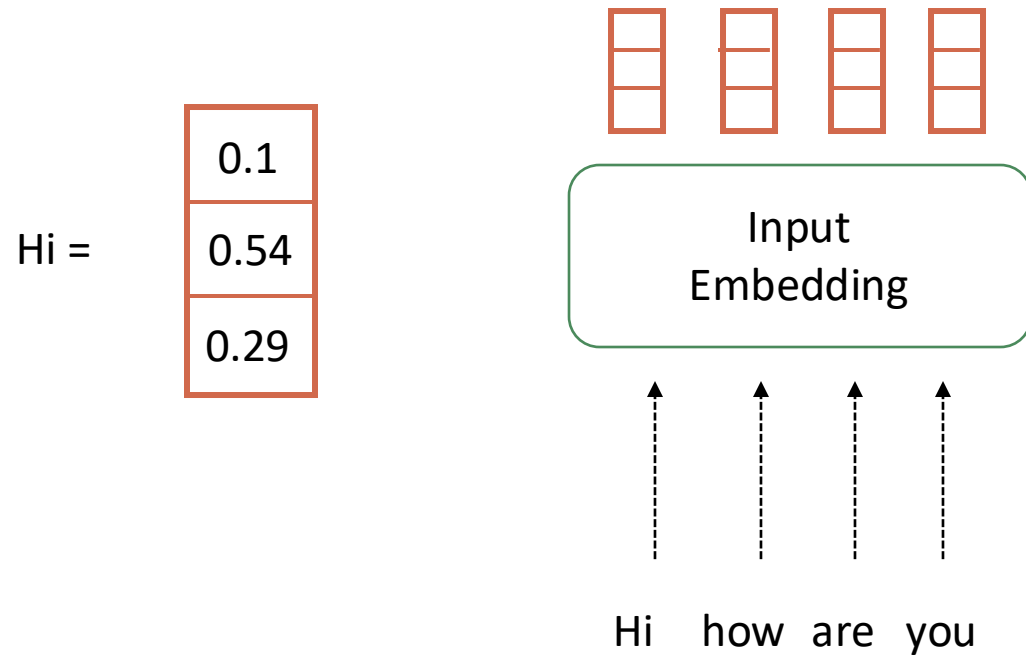
Computer Vision COMP90086 Workshop Week 6

Thi Xuan May Le (May)

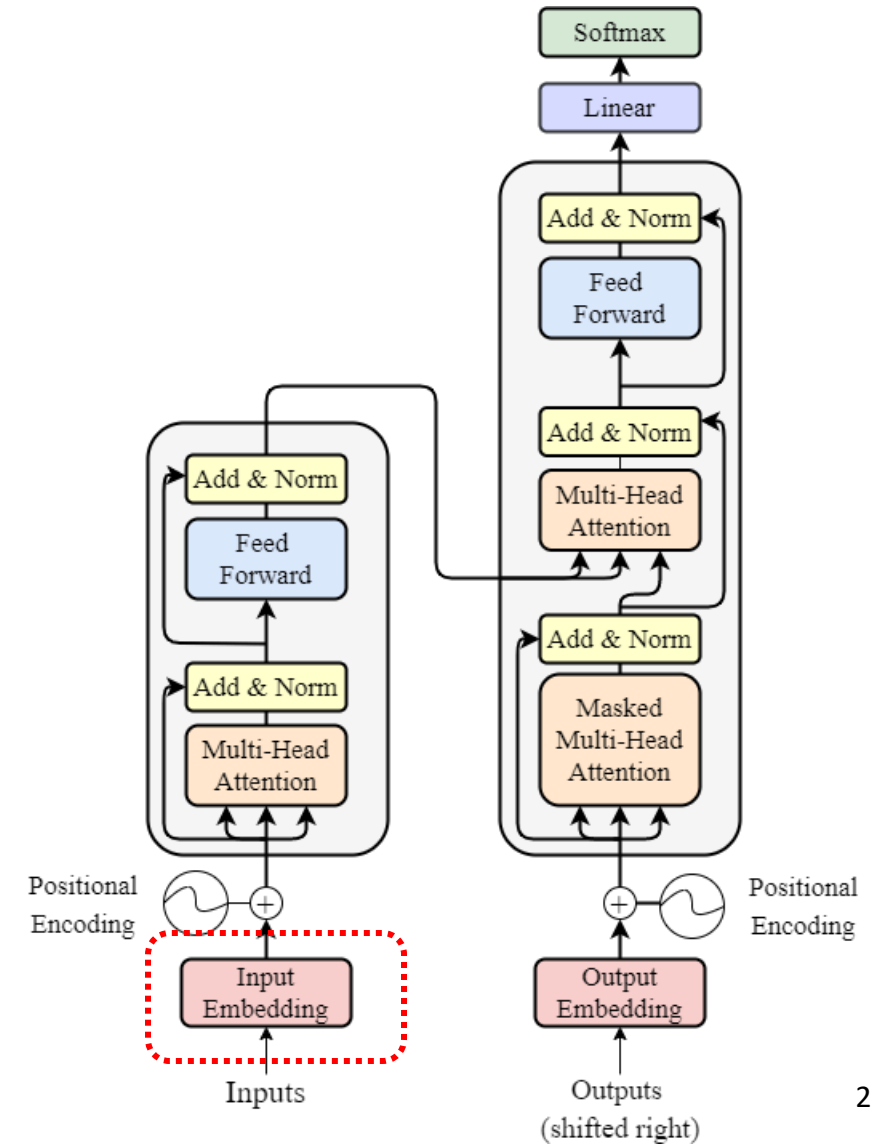
thixuanmay.le@unimelb.edu.au



Transformer Model



1. Input Embedding



Positional Input
Embeddings



Positional
Encoding



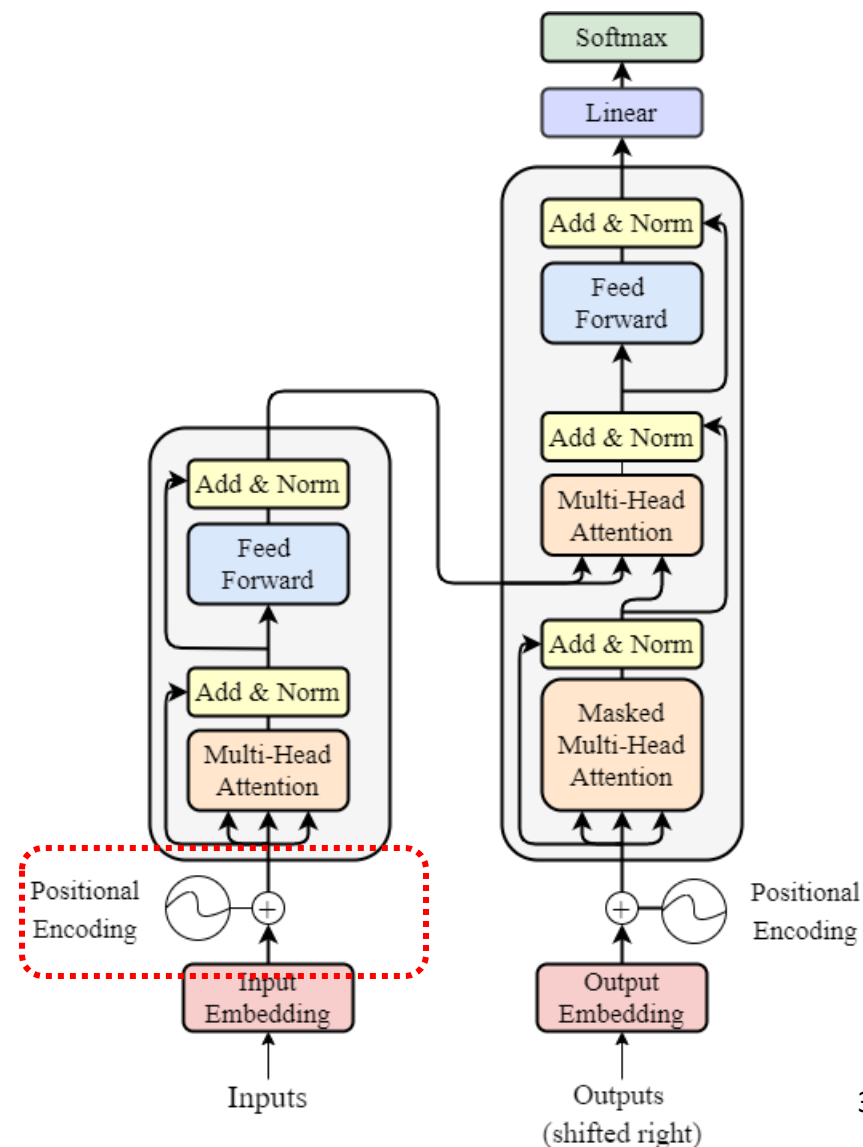
Time Step

1 2 3 4

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

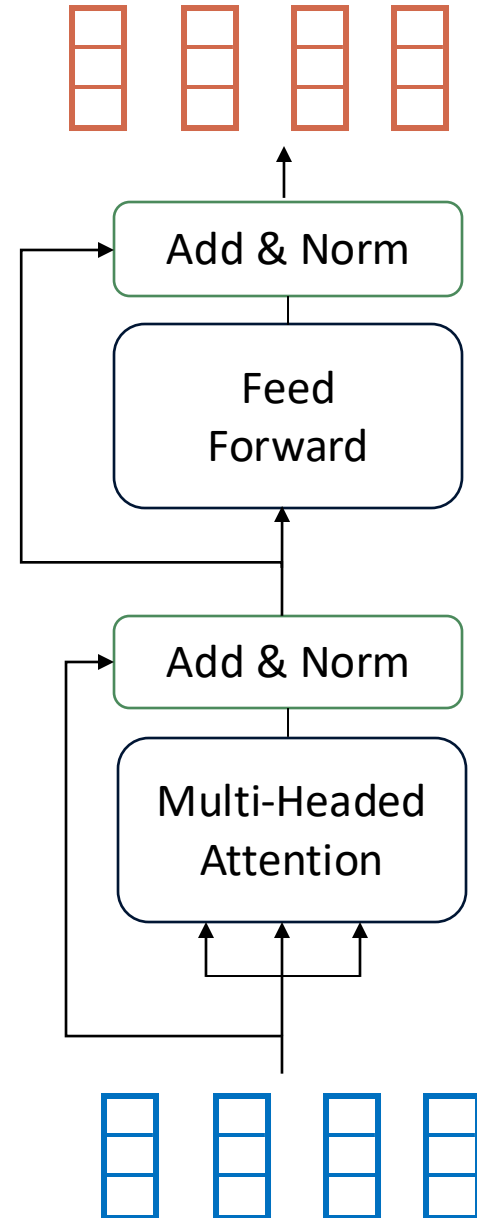
$$PE(sin, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

2. Position Embedding

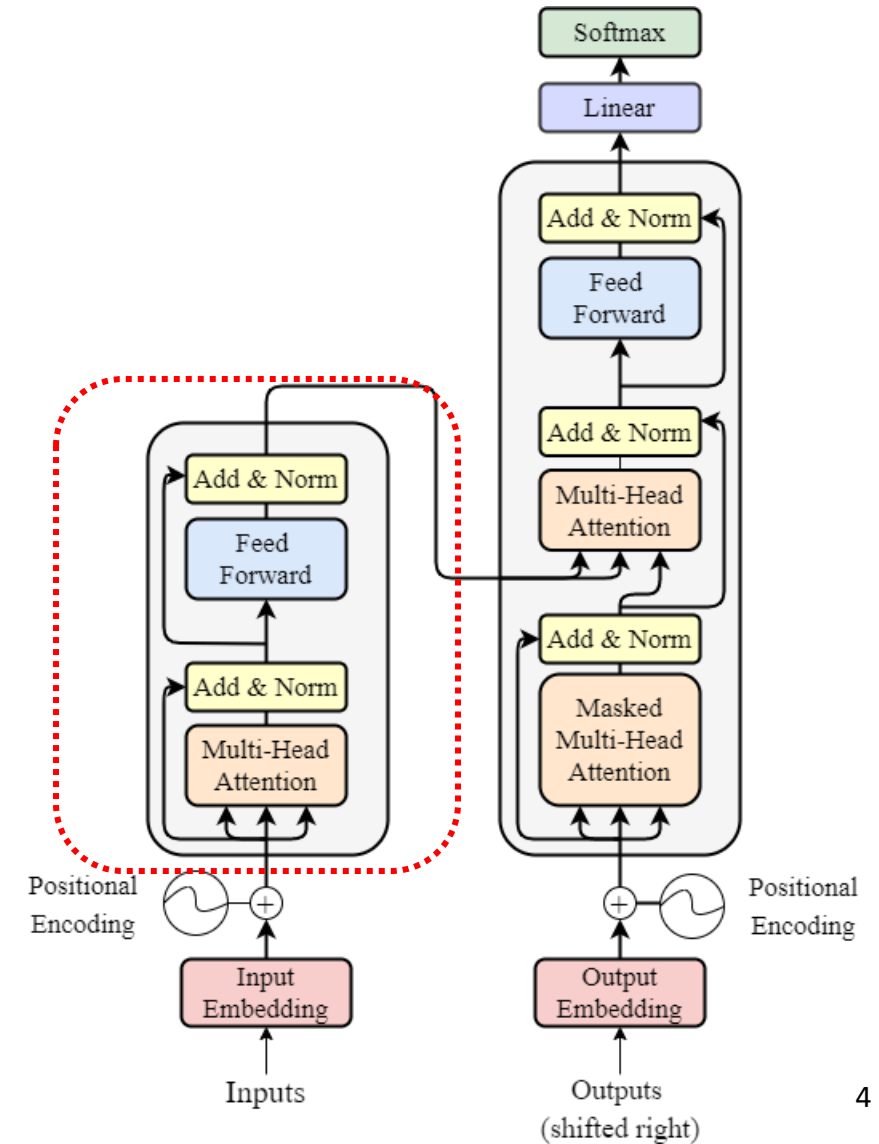


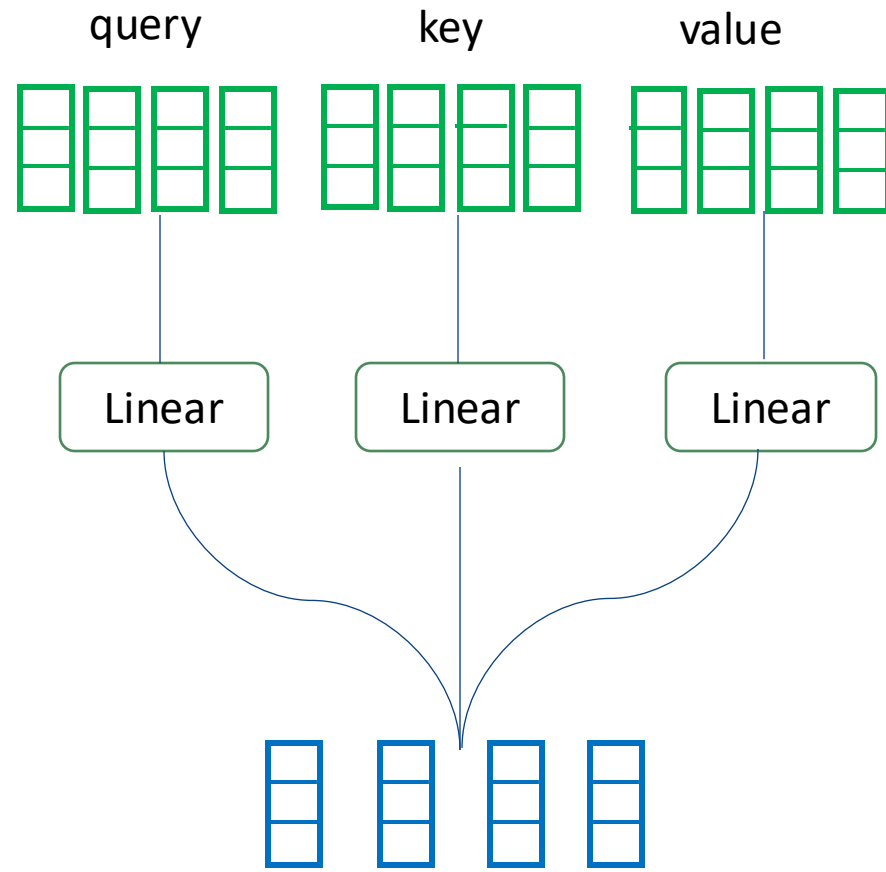
Encoder Input Representation

Positional Input Embeddings

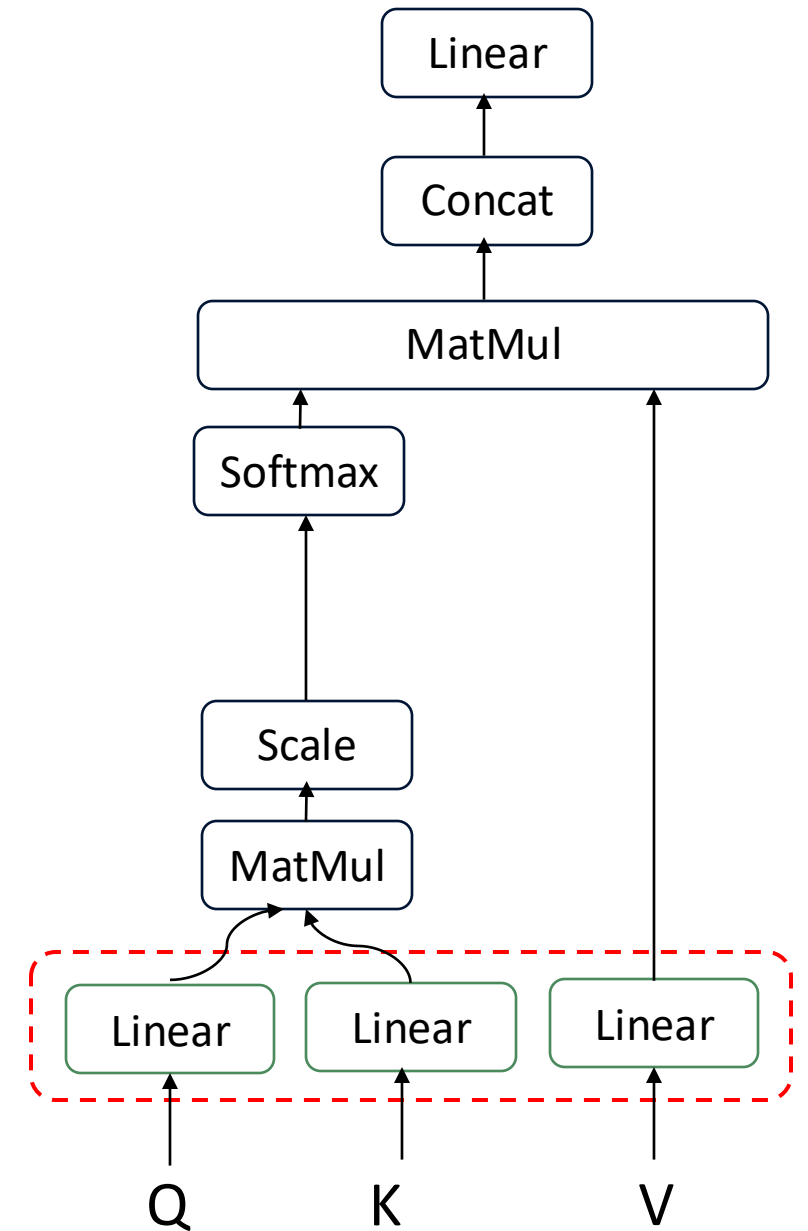


3-4. Encoder Layer



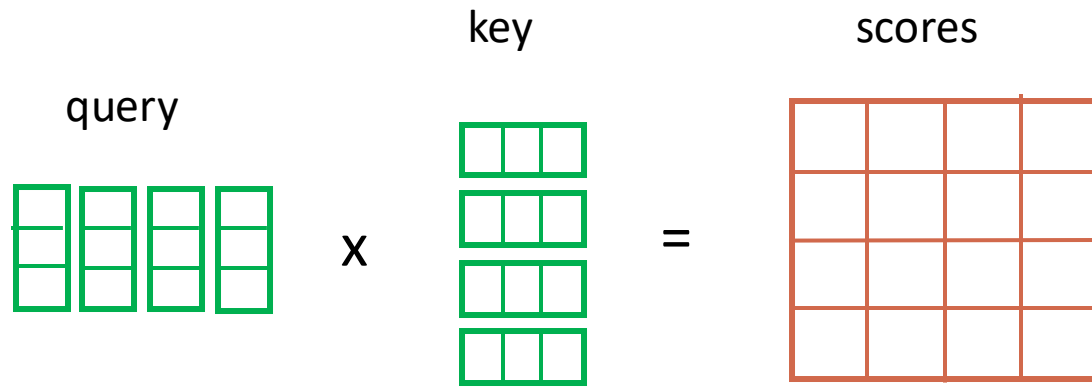


3. Multi-headed Attention



3. Multi-headed Attention

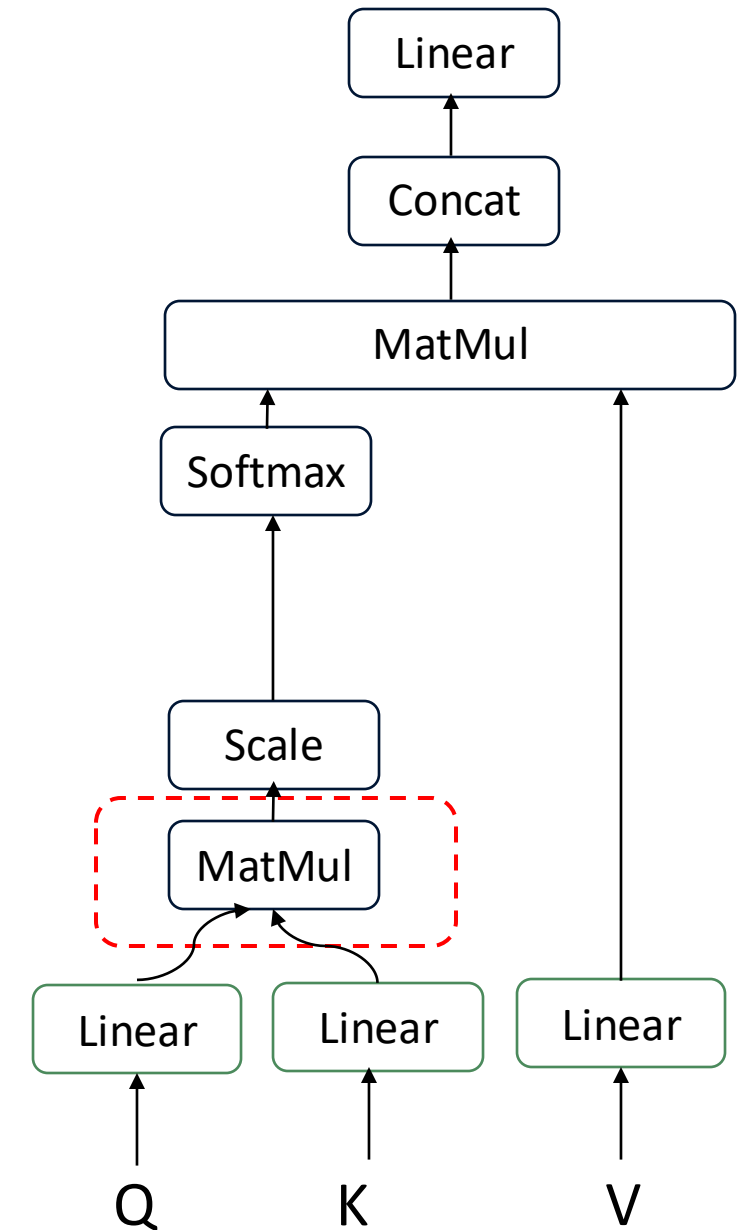
3.1 Self-Attention



Hi how are you

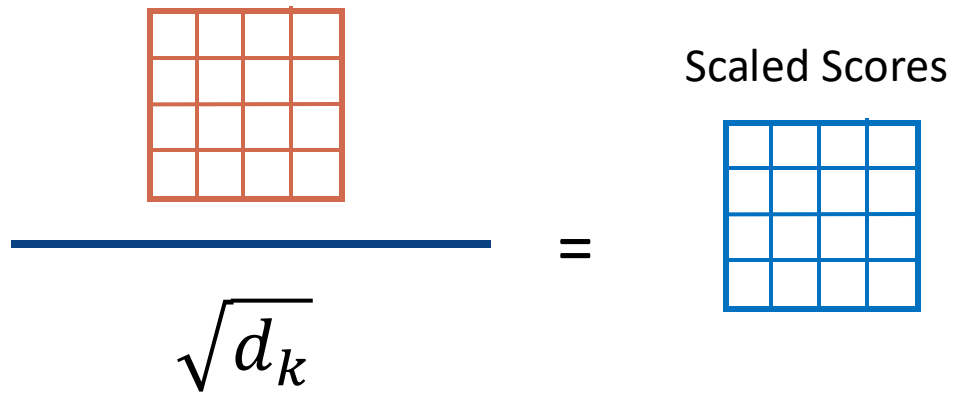
Hi	0.5	0.1	0.5	0.3
how	0.5	0.7	0.7	0.1
are	0.3	0.3	0.1	0.6
you	0.1	0.2	0.7	0.3

value

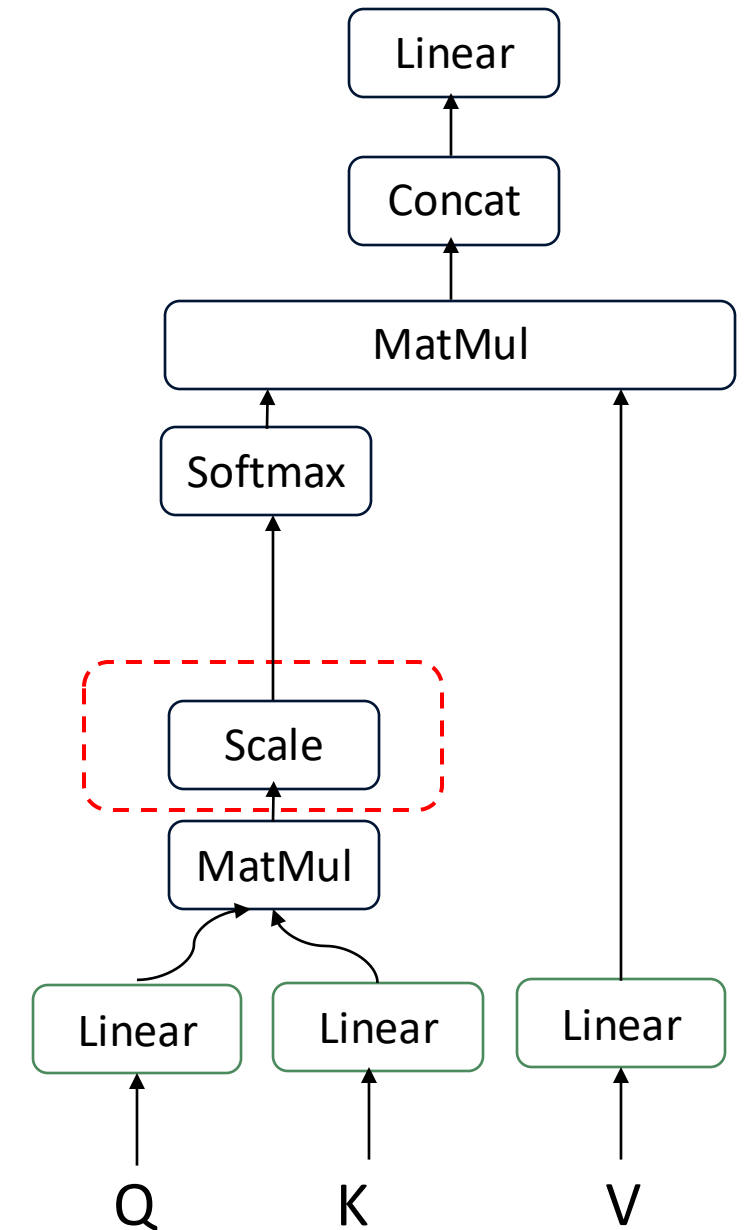



3. Multi-headed Attention

3.1 Self-Attention



Scaled Scores

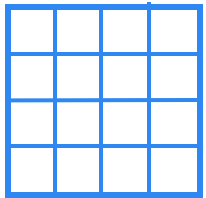


3. Multi-headed Attention

3.1 Self-Attention

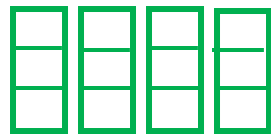
$$\text{Softmax}\left(\begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline \end{array}\right)$$

attention weights



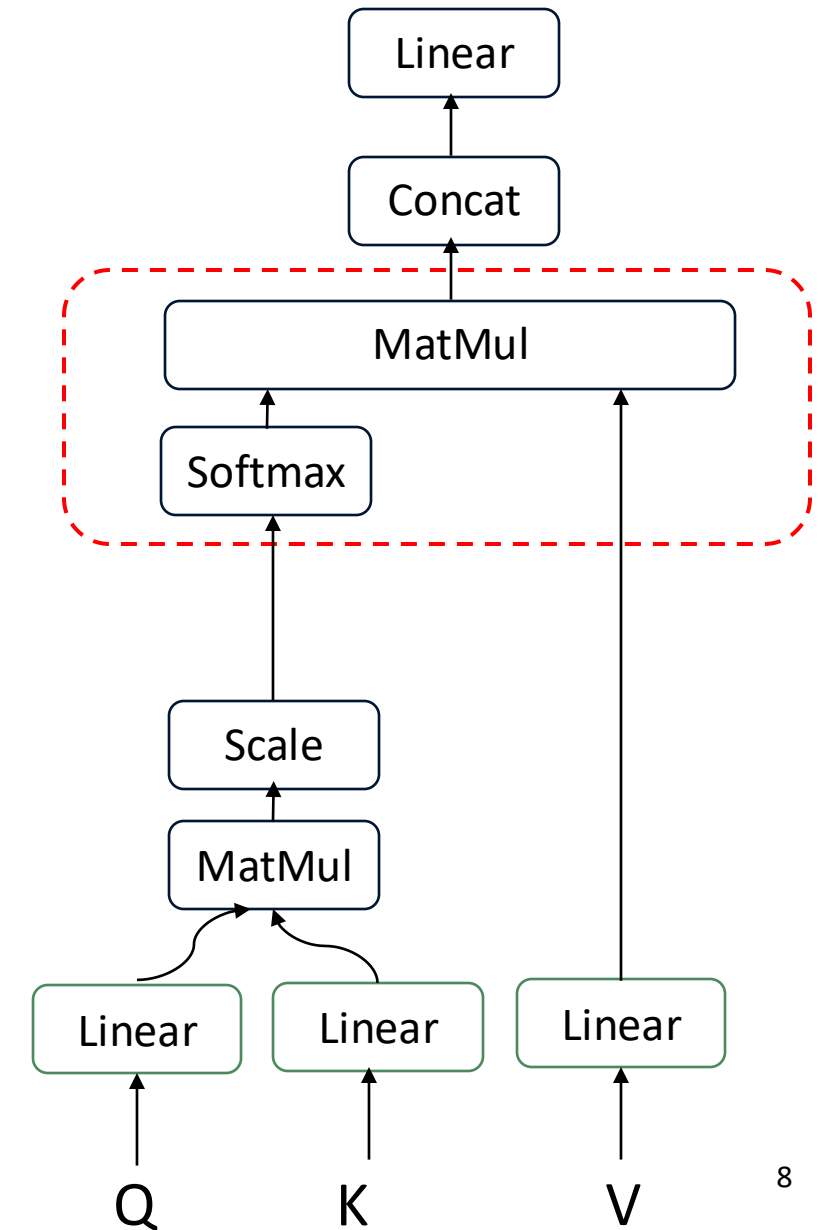
x

value



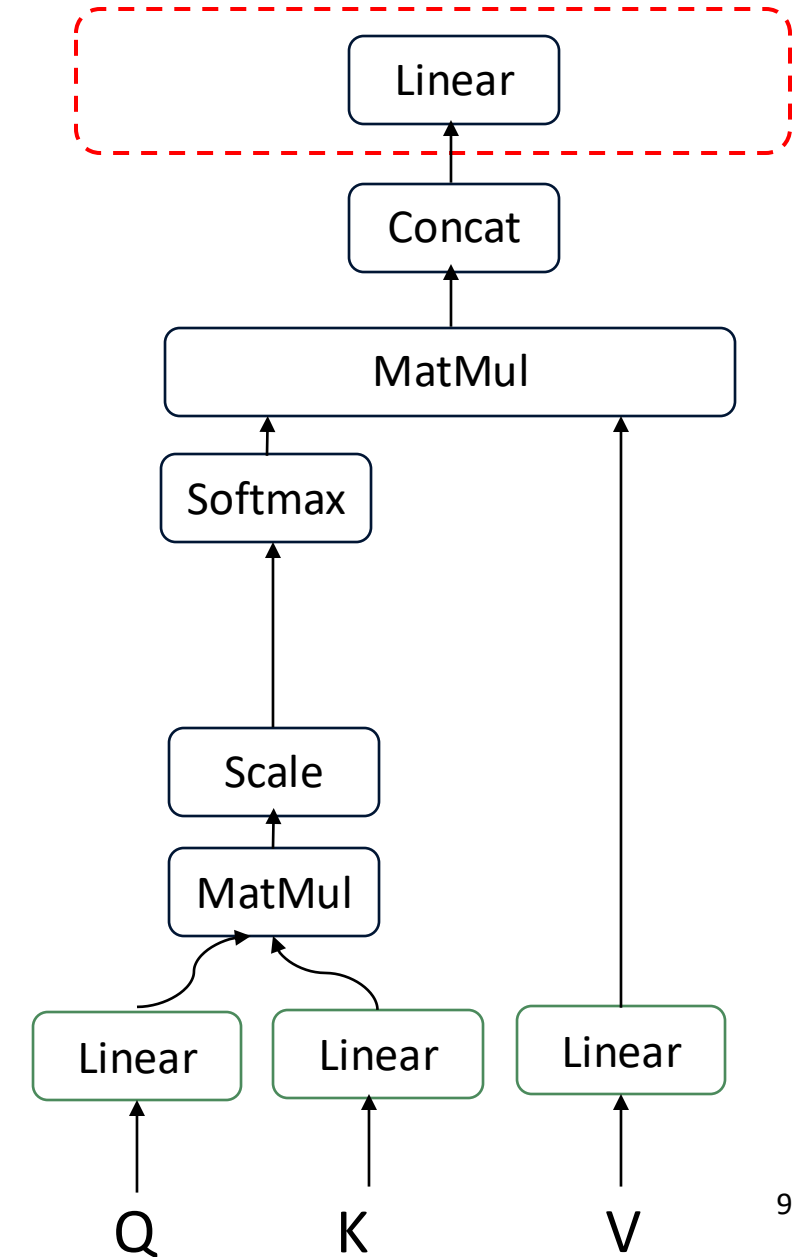
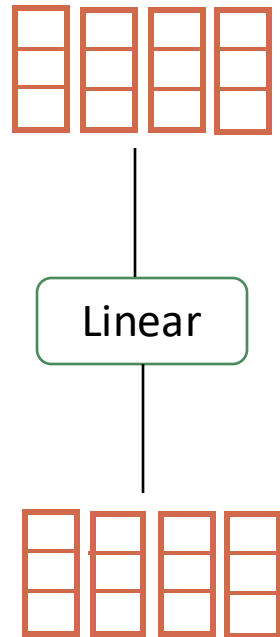
=

output

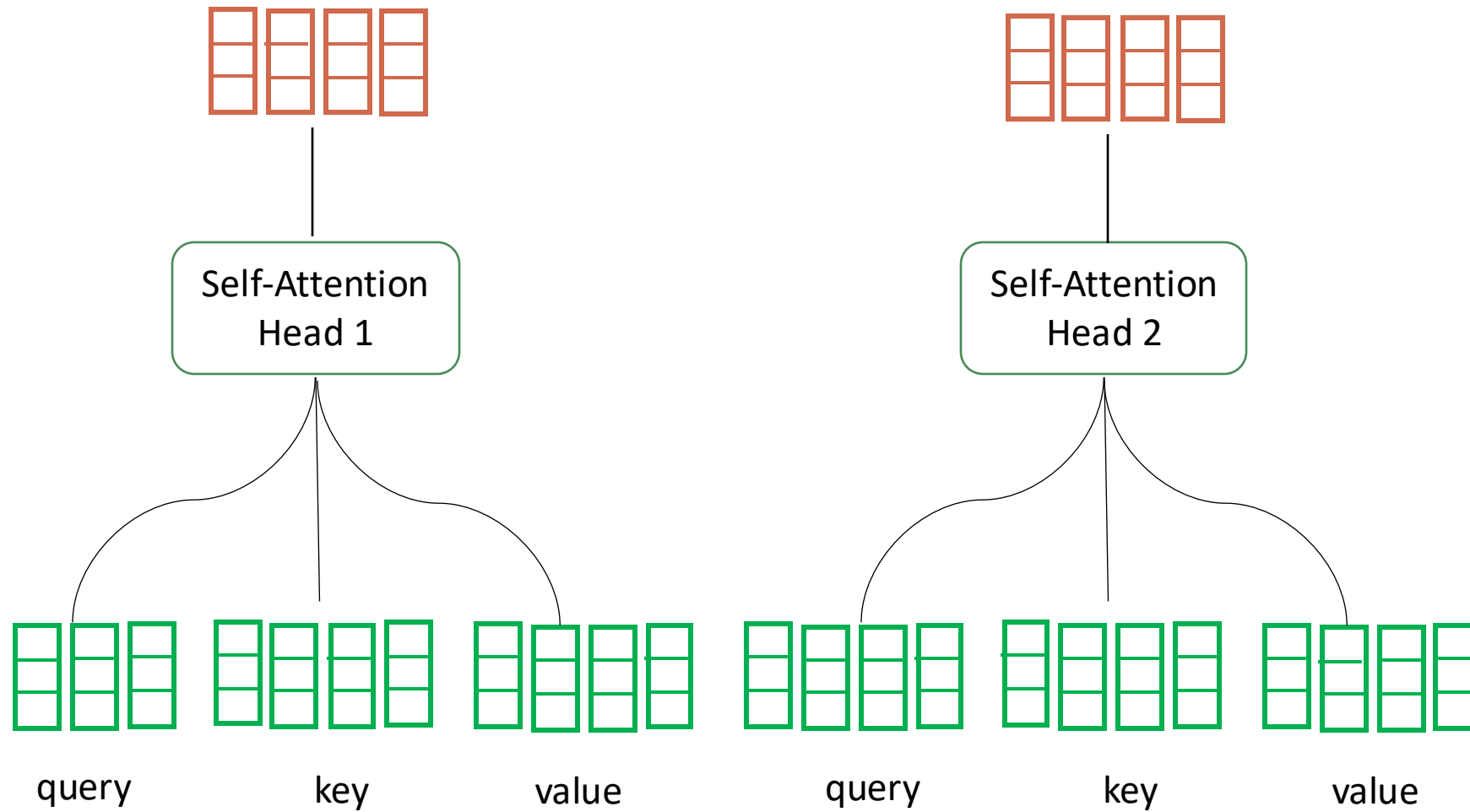


3. Multi-headed Attention

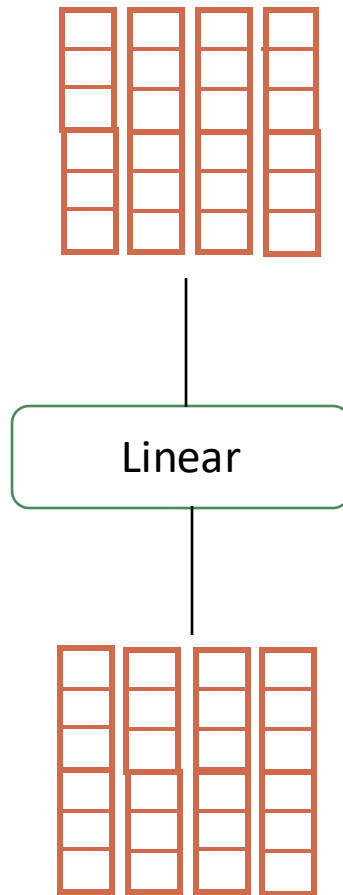
3.1 Self-Attention



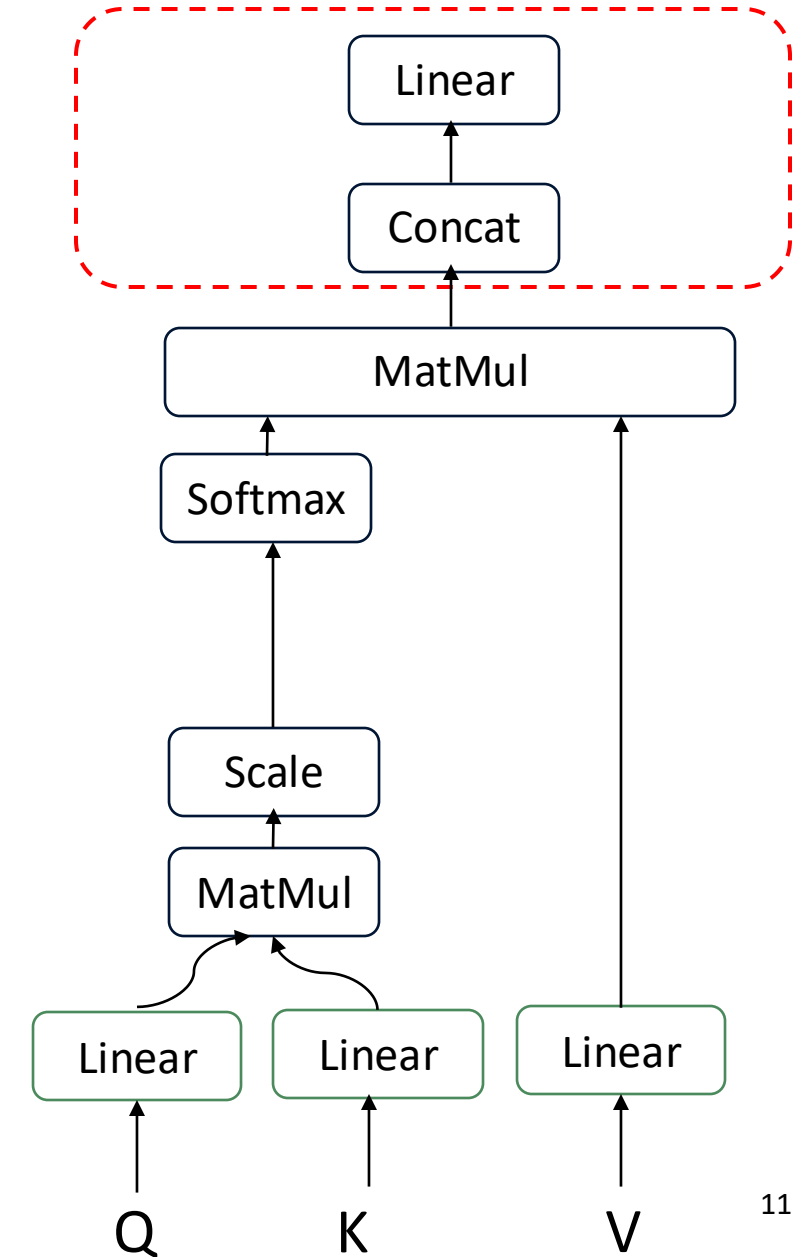
3. Multi-headed Attention



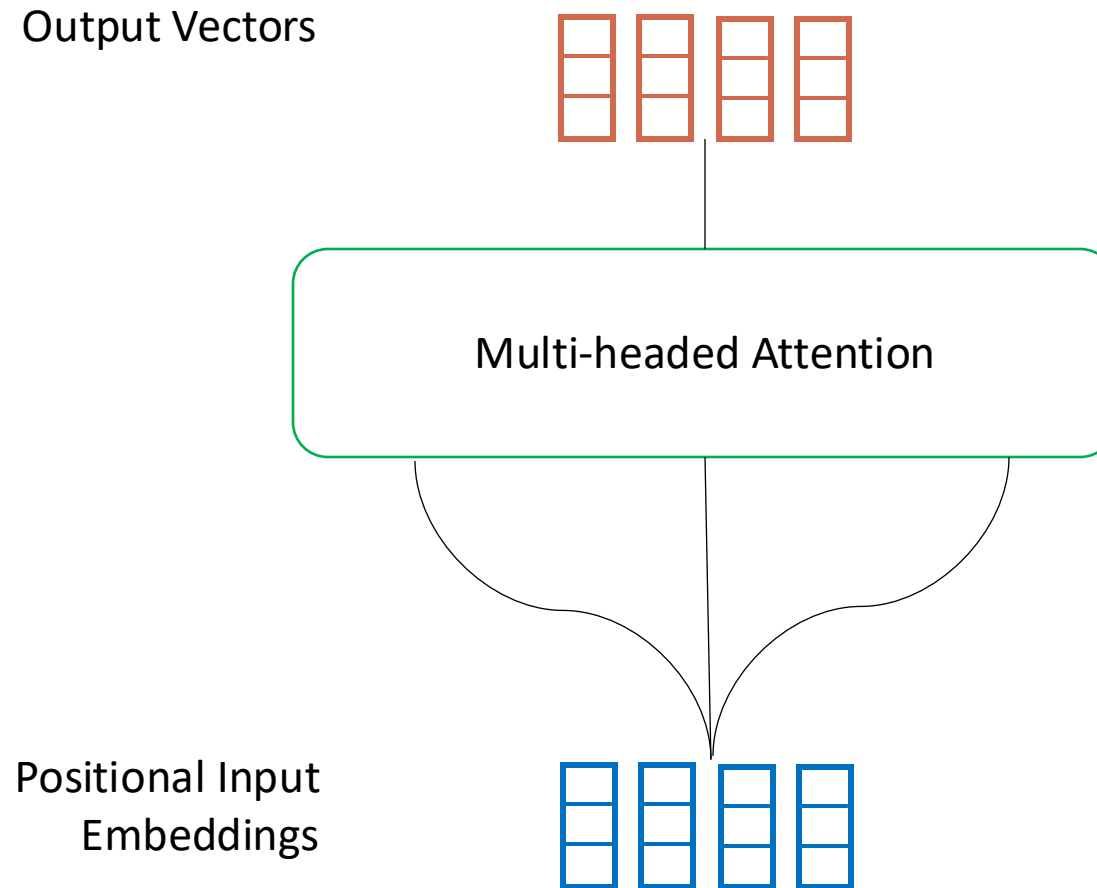
$N = 2$



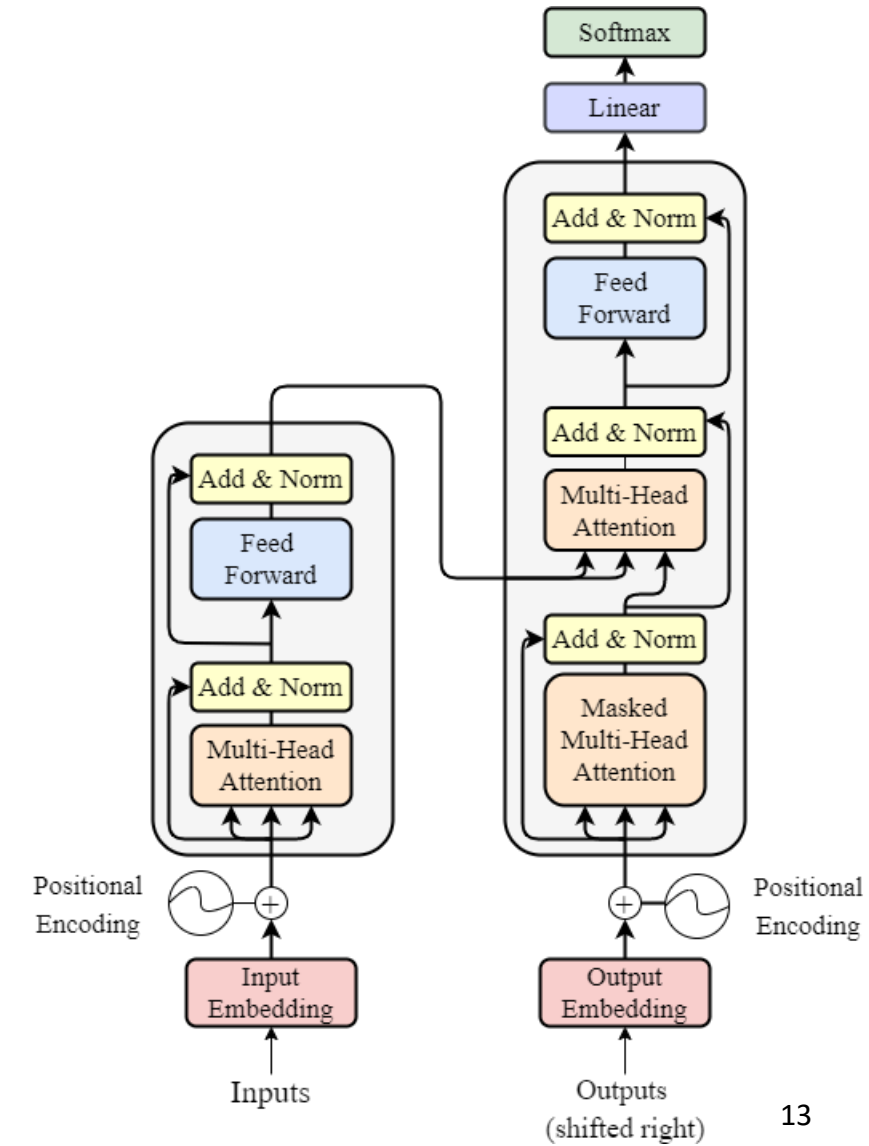
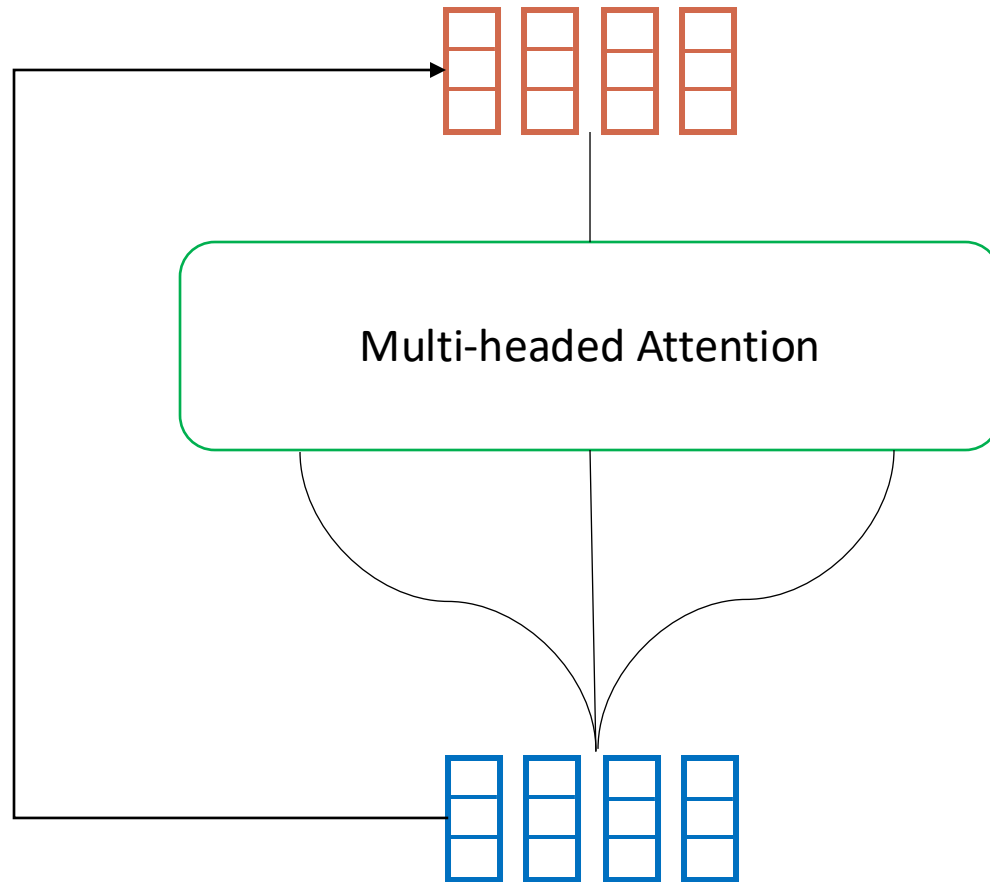
3. Multi-headed Attention

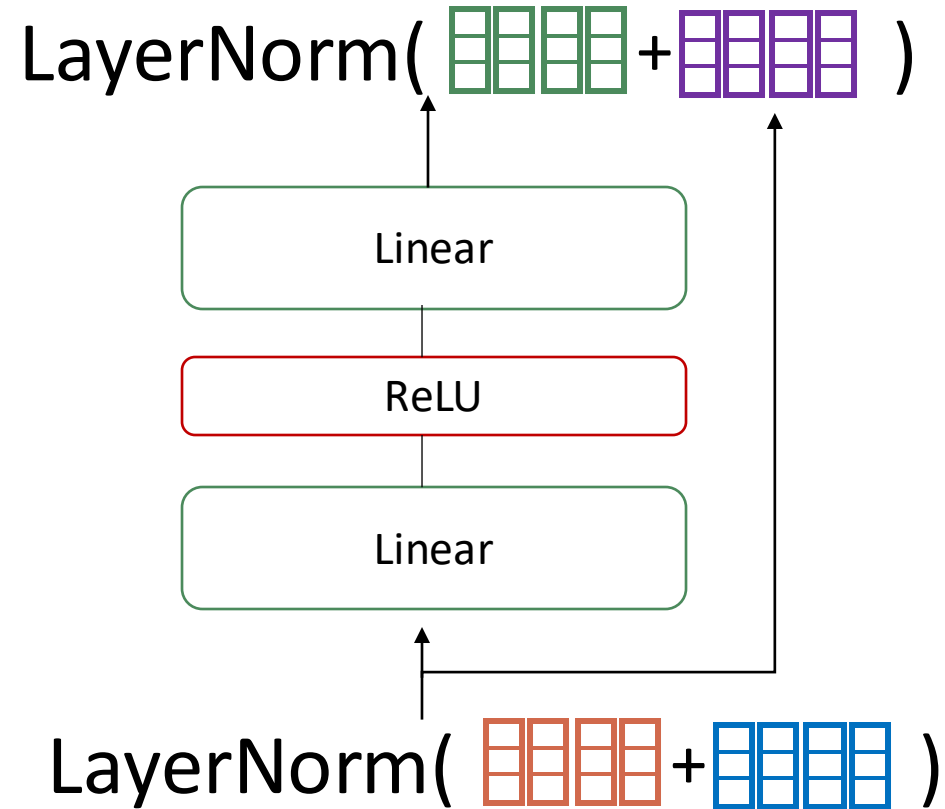


3. Multi-headed Attention

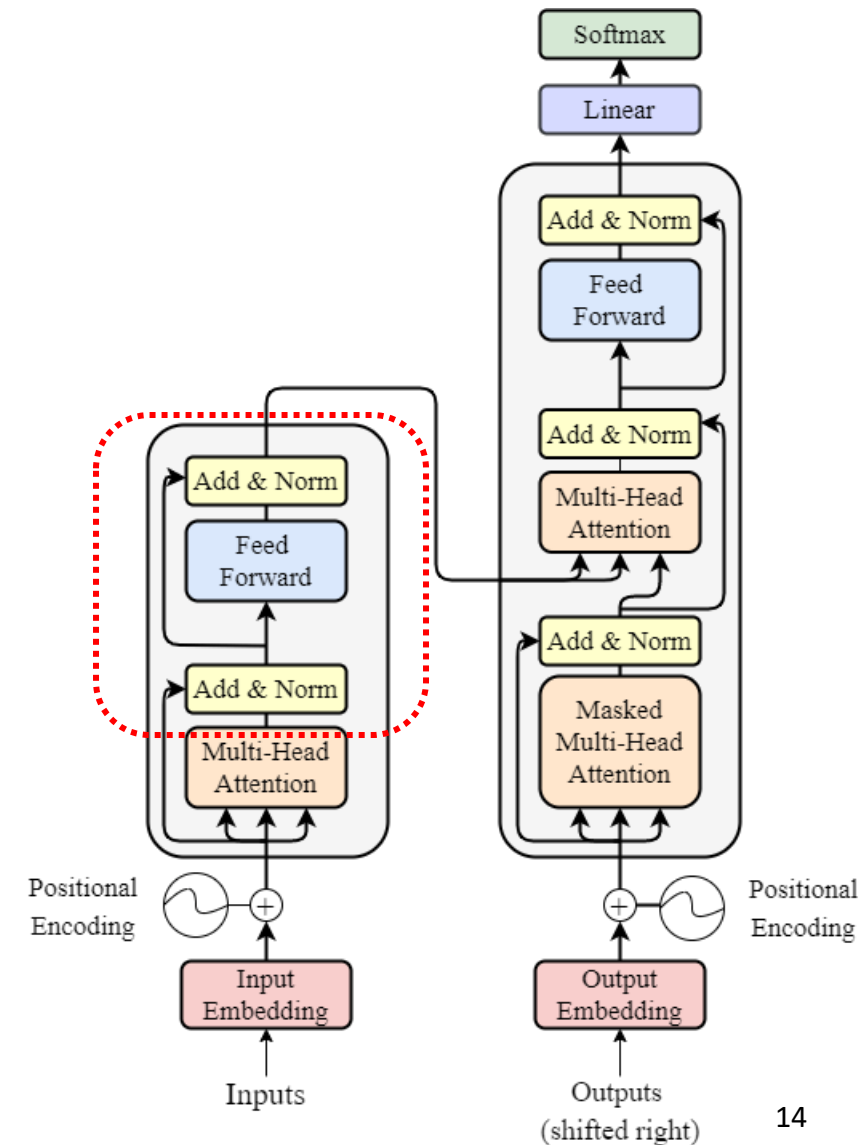


4. Residual Connection, Layer Normalization & Pointwise Feed Forward

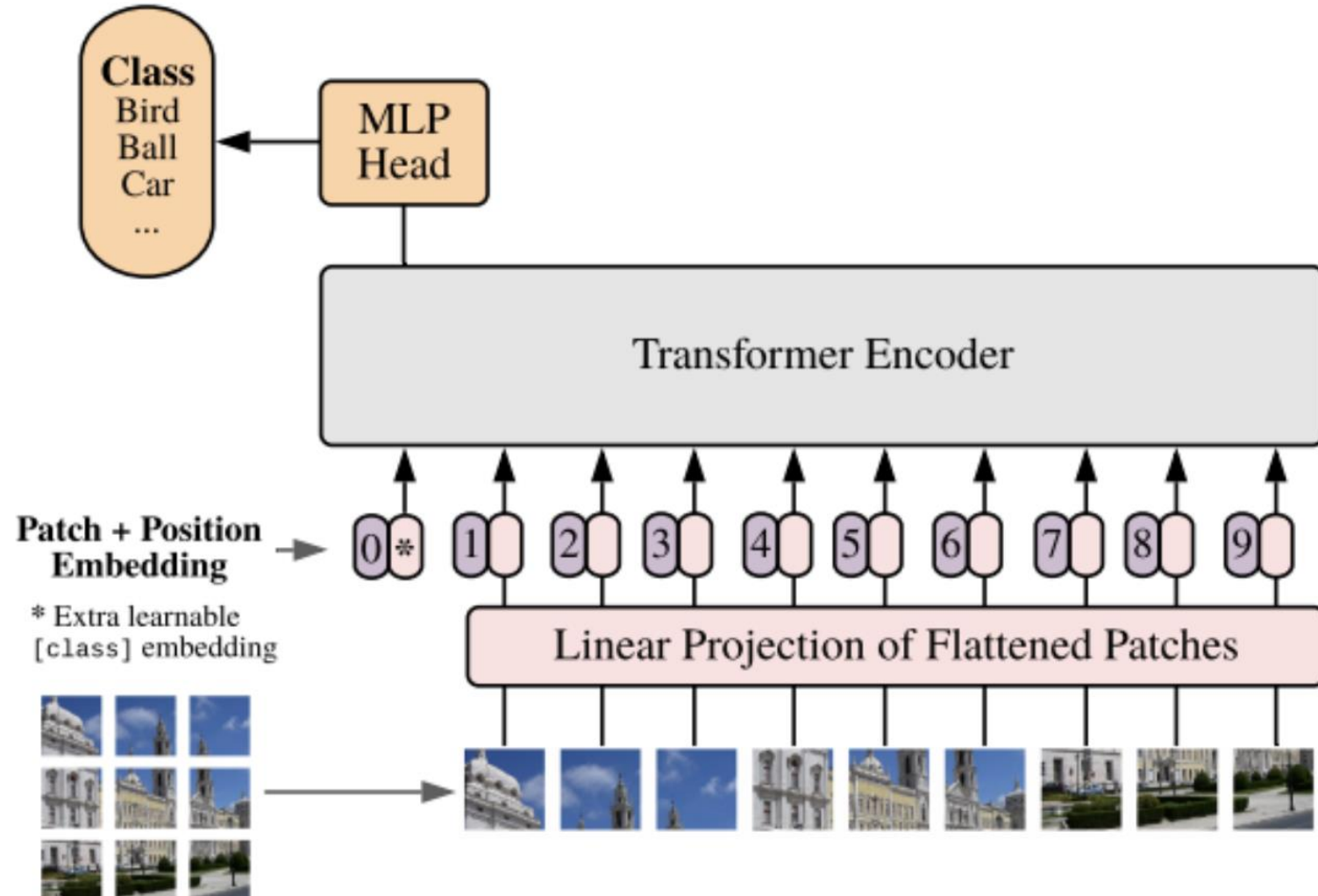




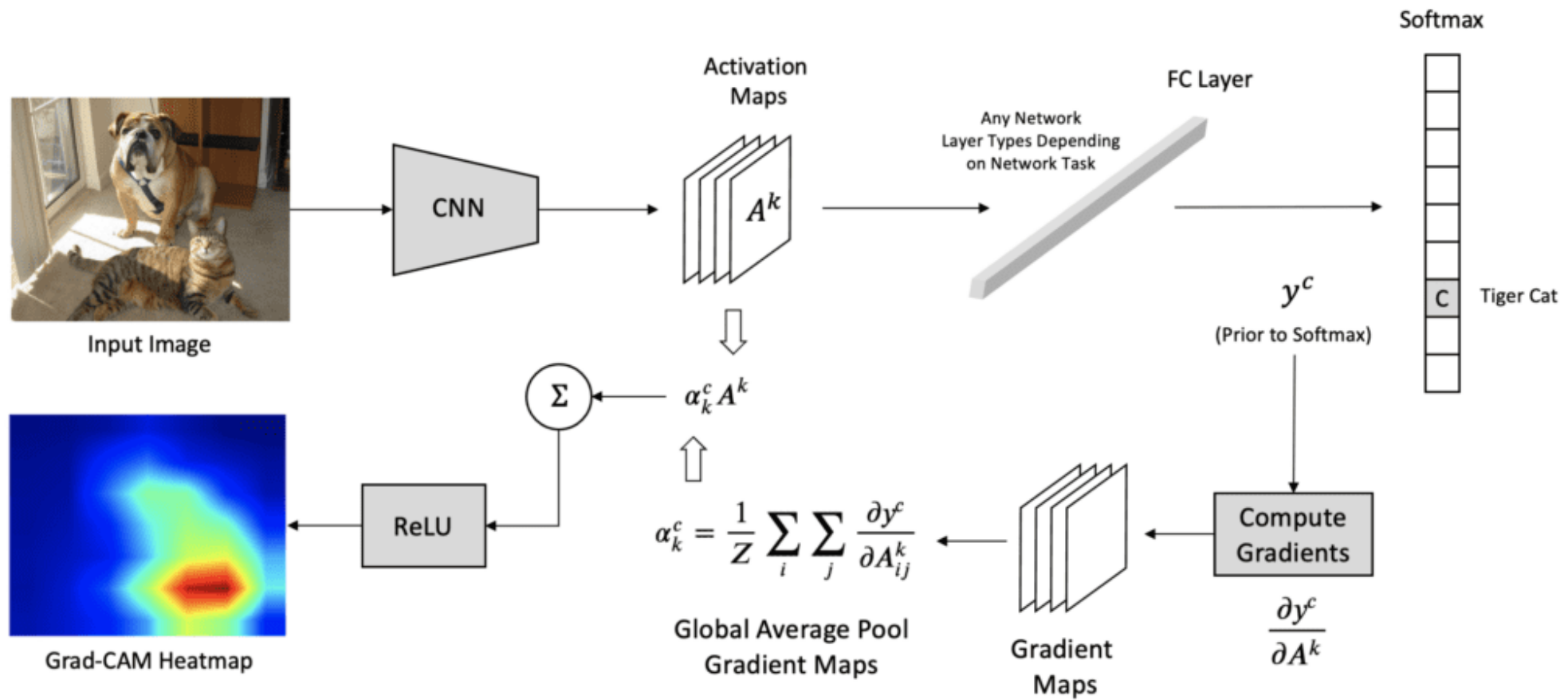
4. Residual Connection, Layer Normalization & Pointwise Feed Forward



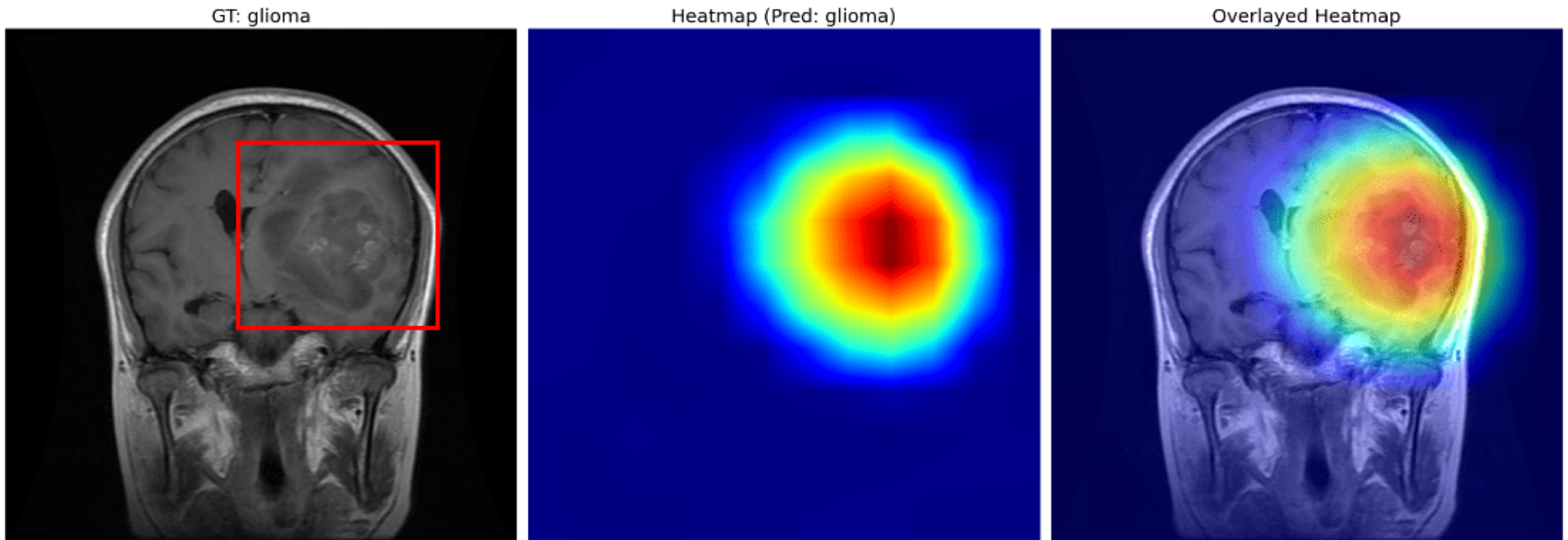
Vision Transformer (ViT)



Grad-CAM for CNNs

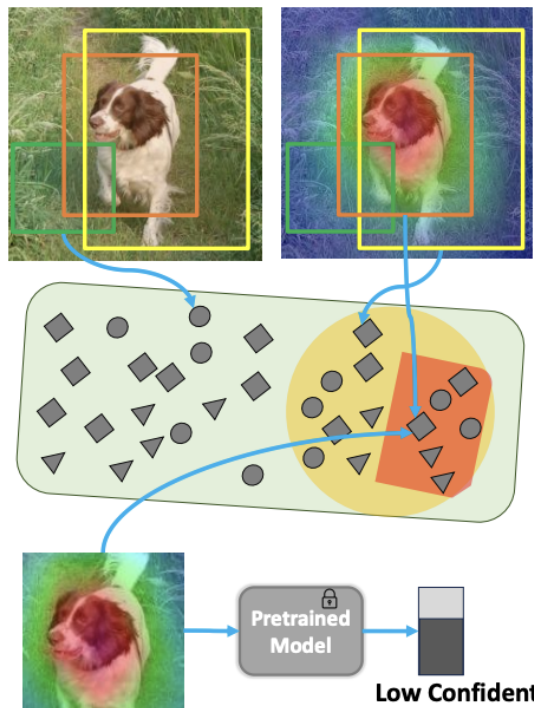


Applications of Grad-CAM in Medical Domain



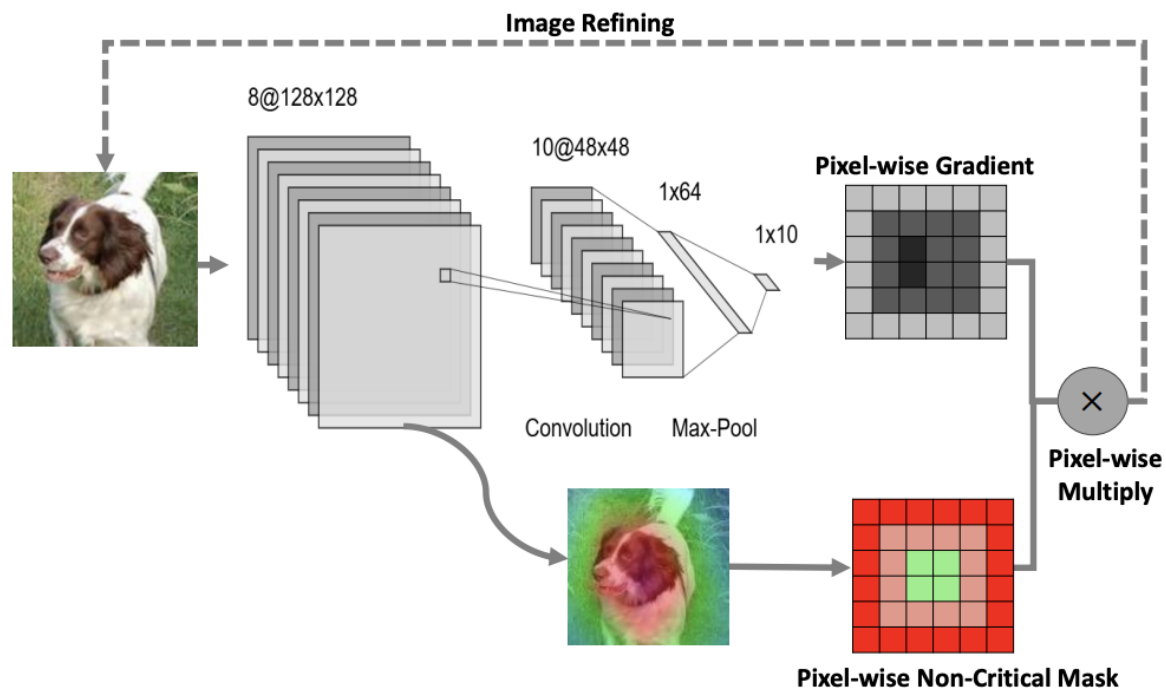
*Image from Google

■ Random Crop ■ Crop with High CAM Ratio
■ Random Crop with High CAM Ratio & Low Confident



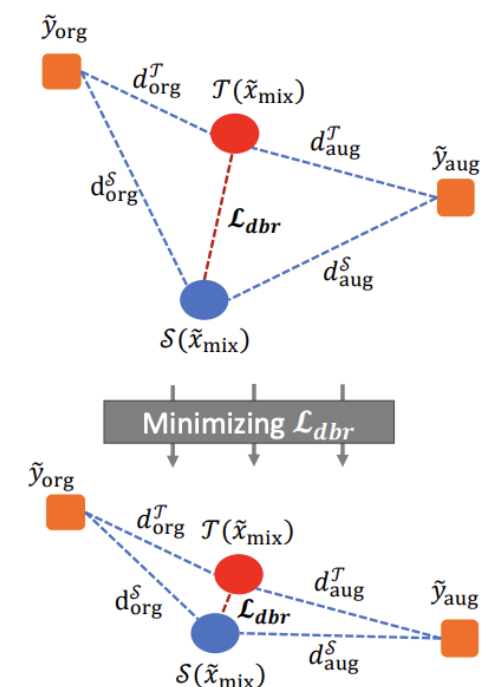
(a) Critical-Based Data Discovery

■ Red in the mask indicates high refine ■ Green in the mask indicates no refine



(b) Non-Critical Region Refinement

■ One-hot vector ● Soft-label



(c) Distance-based Representative



Today's materials

Please access with
UniMelb account.

